

manifold:

We say a topo. space M is an n -manifold iff

$\forall x \in M, \exists U \ni x$ such that $U \cong \mathbb{R}^n \cong B^n := \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$
 "is homeomorphic to"

Colloquially:
 Each point locally looks like $\mathbb{R}^n$.
 → means $\exists$ an open neighborhood such that ...

M is an n -manifold with boundary iff $\forall x \in M \exists U \ni x$ such that

$U \cong \mathbb{R}^n$ or $U \cong \mathbb{H}^n := \{x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_1 \geq 0\}$
 manifold-like pts (interior) boundary $bd(M) = \partial(M) = \text{set of points s.t. } \exists U \ni p \cong \mathbb{H}^n$

mfds need not be connected!

manifolds we know + love:

- $\mathbb{R}^n$ • $\mathbb{RP}^2 = \mathbb{P}^2$
- $T^2 = \text{the torus} = S^1 \times S^1$
- Knots are 1-dim manifolds + links + tangles
- $K =$ 

manifolds w/ boundary:

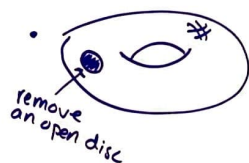
- $\mathbb{H}^n$ 
- 
- a path with end $\neq$ start 
- Möbius strip 

manifolds we know + love:

- $\mathbb{R}^n$ • $\mathbb{RP}^2 = \mathbb{P}^2$
- $\mathbb{T}^2$ = the torus = $S^1 \times S^1$
- Knots are 1-dim manifolds + links + tangles
- $K =$ 

manifolds w/ boundary:

- H^n
-  3-fld w/ bdy



- a path with end $\neq$ start
- Möbius strip  = 

Surface := a 2-manifold in $\mathbb{R}^n$

Orientability of surfaces:

- $M =$ the surface
- pick a point p + assign it an orientation $\rightarrow$ think of RHR giving us a normal direction

- if $\forall$ closed curves

$\gamma: S^1 \rightarrow M$ such that $\gamma(0) = p$

the orientation moving along this path does not reverse.

(Such a path would be called an orientation-reversing path).

Q 1. Find an orientation-reversing curve in the Möbius strip

2. Pick a closed curve on $\mathbb{T}^2$ and verify that it is orientation-preserving.

3. Is there an orientation-preserving curve on Mö

$$\mathbb{T}^2 \cong \mathbb{R}^2 / \sim$$

$$a \sim b \Leftrightarrow a = b + c, \quad c \in \mathbb{Z}^2$$

$$\cong \text{  }$$

manifold in $\mathbb{R}^n$

surfaces:

it an

$\mathbb{R}^n$ giving us a
normal direction

that $\gamma(0) = p$

along this path

called an
path).

[Q] 1. Find an orientation-reversing
curve in the Möbius strip

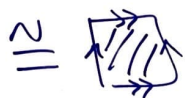
2. Pick a closed curve on
 $\mathbb{T}^2$ and verify that it is
orientation-preserving.

3. Is there an orientation-
preserving curve on Möbius strip?

helpful

$$\mathbb{T}^2 \cong \mathbb{R}^2 / \sim$$

$$a \sim b \iff a = b + c, \\ c \in \mathbb{Z}^2$$



A topologic space T is compact if every open cover admits a finite sub cover.

$(T, \mathcal{T})$ is a topo space
 $S \subseteq \mathcal{T}$ is an ^{open} cover of T
 if $\forall p \in T, \exists S \in \mathcal{S}$ such that $p \in S$

$\mathbb{R}^d$: Compact $\iff$ closed + bounded
 [Q] Prove this!

Compact


not compact

$\mathbb{R}^d \leftarrow (\cdot) \rightarrow$
 $\{(x-1, x+1) \mid x \in \mathbb{R}^d\}$
 No finite subcover!

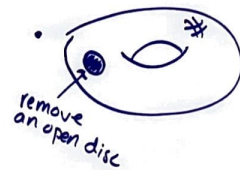
Compact is nice b/c
 inf/sup become min/max
 and more...

manifolds we know + love:

- $\mathbb{R}^n$ • $\mathbb{RP}^2 = \mathbb{P}^2$
- $T^2 =$ the torus $= S^1 \times S^1$
- Knots are 1-dim manifolds + links + tangles
- $K =$ 

manifolds w/ boundary:

- H^n • 



- a path with end $\neq$ start
- Möbius strip  = 

M is orientable if $\left\{ \begin{array}{l} \text{M} = \text{pic} \\ \text{on} \end{array} \right.$